

Using the full-position Fourier gauge and a two-sublattice Néel convention, the calculation proceeds as

$$\mathcal{H}_{\text{micro}} \rightarrow (A_{\mathbf{k}}, B_{\mathbf{k}}) \rightarrow \mathcal{H}_{\text{BdG}}(\mathbf{k}) \rightarrow \omega_n(\mathbf{k}) \rightarrow H_{\text{eff}}^{(2)}(\mathbf{Q}, B) \rightarrow B^*(\mathbf{Q}), C(B).$$

Here $\lambda_{\text{AM}}(\mathbf{k})$ is the microscopic exchange form factor, $\Lambda_{\mathbf{Q}}$ is the signed diagonal detuning in the target positive-frequency doublet, and $D_{\mathbf{Q}}$ is the gauge-invariant transverse mixing obtained by projecting the full DDI operator.

1. Lattice, magnetic order, and local frames

Take the tetragonal lattice vectors

$$\mathbf{a}_1 = (a, 0, 0), \quad \mathbf{a}_2 = (0, b, 0), \quad \mathbf{a}_3 = (0, 0, c), \quad (1)$$

and the two Mn sublattice positions

$$\mathbf{r}_A = (0, 0, 0), \quad \mathbf{r}_B = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right). \quad (2)$$

The ordered state is $A \uparrow, B \downarrow$ with the Néel vector along the crystallographic c axis. To use a common Holstein–Primakoff (HP) expansion on both sublattices, define

$$R_A = I_3, \quad R_B = \text{diag}(-1, 1, -1), \quad (3)$$

$$\mathbf{S}_{i\mu} = R_{\mu} \tilde{\mathbf{S}}_{i\mu}, \quad \mu = A, B. \quad (4)$$

Both ordered moments then point along local $+\tilde{z}$.

In the full-position Fourier gauge, the displacement of a bond from sublattice μ to ν is

$$\boldsymbol{\delta} = \mathbf{R} + \mathbf{r}_{\nu} - \mathbf{r}_{\mu}, \quad (5)$$

with Fourier phase $e^{i\mathbf{k} \cdot \boldsymbol{\delta}}$. Equivalently,

$$\tilde{c}_{\mu, \mathbf{R}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} c_{\mu, \mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{R} + \mathbf{r}_{\mu})}. \quad (6)$$

This is the full-position gauge, with the basis positions included in the bond displacement $\boldsymbol{\delta}$.

2. Microscopic Hamiltonian

The complete model is

$$\mathcal{H} = \mathcal{H}_{\text{ex}} + \mathcal{H}_{\text{SIA}} + \mathcal{H}_{\text{dip}} + \mathcal{H}_Z. \quad (7)$$

The individual terms are

$$\mathcal{H}_{\text{ex}} = \sum_{(i\mu, j\nu) \in \mathcal{B}_{\text{ex}}} J_{i\mu, j\nu} \mathbf{S}_{i\mu} \cdot \mathbf{S}_{j\nu}, \quad (8)$$

$$\mathcal{H}_{\text{SIA}} = \sum_{i\mu} \mathbf{S}_{i\mu}^T \mathbf{A}_{\mu} \mathbf{S}_{i\mu}, \quad (9)$$

$$\mathcal{H}_{\text{dip}} = C_{\text{dip}} \sum_{i\mu < j\nu} \left[\frac{\mathbf{S}_{i\mu} \cdot \mathbf{S}_{j\nu}}{r^3} - 3 \frac{(\mathbf{S}_{i\mu} \cdot \mathbf{r})(\mathbf{S}_{j\nu} \cdot \mathbf{r})}{r^5} \right], \quad (10)$$

$$\mathcal{H}_Z = -g\mu_B B \sum_{i\mu} S_{i\mu}^z, \quad B \parallel c, \quad (11)$$

where $\mathbf{r} = \mathbf{r}_{j\nu} - \mathbf{r}_{i\mu}$ and

$$C_{\text{dip}} = \frac{\mu_0}{4\pi} (g\mu_B)^2. \quad (12)$$

The unit conversion is $\mu_0\mu_B^2/(4\pi \text{Å}^3) = 0.0536815 \text{ meV}$.

3. HP expansion and generic BdG rules

In each sublattice local frame,

$$\tilde{S}^z = S - c^\dagger c, \quad \tilde{S}^+ = \sqrt{2S} c, \quad \tilde{S}^- = \sqrt{2S} c^\dagger, \quad (13)$$

or

$$\tilde{S}^x = \sqrt{\frac{S}{2}} (c + c^\dagger), \quad \tilde{S}^y = -i\sqrt{\frac{S}{2}} (c - c^\dagger). \quad (14)$$

Terms are retained through quadratic order, $O(S)$ in LSWT.

3.1 Generic bilinear tensor interaction

For a global tensor coupling

$$\mathbf{S}_{i\mu}^\top \mathbf{K}_{\mu\nu}^{\text{glob}} \mathbf{S}_{j\nu}, \quad (15)$$

first rotate it to the local frames,

$$\mathbf{K}_{\mu\nu}^{\text{loc}} = \mathbf{R}_\mu^\top \mathbf{K}_{\mu\nu}^{\text{glob}} \mathbf{R}_\nu. \quad (16)$$

Writing the components as $K_{\alpha\beta}$, the transverse HP coefficients are

$$t_{\mu\nu} = \frac{S}{2} [(K_{xx} + K_{yy}) + i(K_{yx} - K_{xy})], \quad (17)$$

$$p_{\mu\nu} = \frac{S}{2} [(K_{xx} - K_{yy}) + i(K_{xy} + K_{yx})]. \quad (18)$$

The longitudinal contribution is set by K_{zz} ; t enters the normal block A and p the anomalous/pairing block B .

3.2 Same-sublattice isotropic exchange

For a same-sublattice isotropic bond,

$$J\tilde{\mathbf{S}}_{\mathbf{R}} \cdot \tilde{\mathbf{S}}_{\mathbf{R}+\delta}, \quad (19)$$

the quadratic part is

$$JS^2 + JS \left(c_{\mathbf{R}}^\dagger c_{\mathbf{R}+\delta} + c_{\mathbf{R}+\delta}^\dagger c_{\mathbf{R}} - n_{\mathbf{R}} - n_{\mathbf{R}+\delta} \right). \quad (20)$$

After Fourier transformation,

$$\mathcal{H}_{J,\delta}^{(2)} = 2JS \sum_{\mathbf{k}} [\cos(\mathbf{k} \cdot \delta) - 1] c_{\mathbf{k}}^\dagger c_{\mathbf{k}}. \quad (21)$$

Same-sublattice isotropic exchange contributes only to the normal block and generates no pairing.

3.3 Opposite-sublattice isotropic exchange

For an A - B bond,

$$\mathbf{R}_A^\top \mathbf{I} \mathbf{R}_B = \text{diag}(-1, 1, -1), \quad (22)$$

so

$$\mathbf{S}_A \cdot \mathbf{S}_B = -\tilde{S}_A^x \tilde{S}_B^x + \tilde{S}_A^y \tilde{S}_B^y - \tilde{S}_A^z \tilde{S}_B^z. \quad (23)$$

At quadratic order,

$$J\mathbf{S}_A \cdot \mathbf{S}_B = -JS^2 + JS(n_A + n_B) - JS(ab + a^\dagger b^\dagger). \quad (24)$$

An antiferromagnetic inter-sublattice bond gives the same $+JS$ restoring contribution to both diagonal normal elements and generates anomalous pairing.

4. Exchange paths J_1 – J_7 : term-by-term derivation

Define

$$x = k_x a, \quad y = k_y b, \quad z = k_z c. \quad (25)$$

The independent bonds used in the two-sublattice calculation are listed below. Same-sublattice displacements are shown only once, with $-\delta$ supplied by the Hermitian conjugate; all $\pm$ combinations are listed explicitly for the inter-sublattice J_2 and J_5 bonds.

Exchange	Type	Independent displacement δ	Coordination
J_1	same sublattice	$(0, 0, c)$	2
J_2	A - B	$(\pm a/2, \pm b/2, \pm c/2)$	8
J_3	same sublattice	$(a, 0, 0), (0, b, 0)$	4
J_4	same sublattice	$(a, 0, \pm c), (0, b, \pm c)$	8
J_5	A - B	$(\pm a/2, \pm b/2, \pm 3c/2)$	8
J_6	same sublattice	$(0, 0, 2c)$	2
J_{7a}, J_{7b}	same sublattice	$(a, -b, 0), (a, b, 0)$ with labels interchanged on B	2 + 2

4.1 J_1 : same-sublattice exchange along c

$$A^{(1)}(\mathbf{k}) = 2SJ_1(\cos z - 1). \quad (26)$$

The contribution is identical on A and B .

4.2 J_2 : nearest inter-sublattice body-diagonal exchange

The structure factor is

$$\gamma_2(\mathbf{k}) = \sum_{s_x, s_y, s_z = \pm 1} e^{\frac{i}{2}(s_x x + s_y y + s_z z)} \quad (27)$$

$$= 8 \cos \frac{x}{2} \cos \frac{y}{2} \cos \frac{z}{2}. \quad (28)$$

Each inter-sublattice bond contributes a diagonal restoring field $+SJ_2$, giving

$$A_{AA}^{(2)} = A_{BB}^{(2)} = 8SJ_2, \quad (29)$$

while the pairing term is

$$B_{AB}^{(2)} = B_{BA}^{(2)} = -8SJ_2 \cos \frac{x}{2} \cos \frac{y}{2} \cos \frac{z}{2}. \quad (30)$$

4.3 J_3 : in-plane axial same-sublattice exchange

$$A^{(3)}(\mathbf{k}) = 2SJ_3(\cos x + \cos y - 2). \quad (31)$$

4.4 J_4 : mixed in-plane and c -axis same-sublattice exchange

$$A^{(4)}(\mathbf{k}) = 2SJ_4 [\cos(x+z) + \cos(x-z) + \cos(y+z) + \cos(y-z) - 4] \quad (32)$$

$$= 4SJ_4 [(\cos x + \cos y) \cos z - 2]. \quad (33)$$

4.5 J_5 : longer-range inter-sublattice exchange

$$\gamma_5(\mathbf{k}) = \sum_{s_x, s_y, s_z = \pm 1} e^{i(s_x x/2 + s_y y/2 + 3s_z z/2)} \quad (34)$$

$$= 8 \cos \frac{x}{2} \cos \frac{y}{2} \cos \frac{3z}{2}, \quad (35)$$

which gives

$$A_{AA}^{(5)} = A_{BB}^{(5)} = 8SJ_5, \quad (36)$$

$$B_{AB}^{(5)} = B_{BA}^{(5)} = -8SJ_5 \cos \frac{x}{2} \cos \frac{y}{2} \cos \frac{3z}{2}. \quad (37)$$

4.6 J_6 : second same-sublattice distance along c

$$A^{(6)}(\mathbf{k}) = 2SJ_6(\cos 2z - 1). \quad (38)$$

4.7 J_{7a}, J_{7b} : altermagnetic diagonal detuning

Define

$$\delta_a = (a, -b, 0), \quad \delta_b = (a, b, 0), \quad \delta J_7 = J_{7b} - J_{7a}, \quad (39)$$

with the J_{7a} and J_{7b} bond labels interchanged on sublattice B . Then

$$A_A^{(7)}(\mathbf{k}) = 2S \{J_{7a} [\cos(x-y) - 1] + J_{7b} [\cos(x+y) - 1]\}, \quad (40)$$

$$A_B^{(7)}(\mathbf{k}) = 2S \{J_{7b} [\cos(x-y) - 1] + J_{7a} [\cos(x+y) - 1]\}. \quad (41)$$

With

$$\bar{J}_7 = \frac{J_{7a} + J_{7b}}{2}, \quad (42)$$

we obtain

$$A_A^{(7)} = 4S\bar{J}_7(\cos x \cos y - 1) - 2S\delta J_7 \sin x \sin y, \quad (43)$$

$$A_B^{(7)} = 4S\bar{J}_7(\cos x \cos y - 1) + 2S\delta J_7 \sin x \sin y. \quad (44)$$

Subtracting the two expressions gives

$$A_A^{(7)} - A_B^{(7)} = -4S\delta J_7 \sin x \sin y. \quad (45)$$

Define

$$\lambda_{AM}(\mathbf{k}) = 4S\delta J_7 \sin(k_x a) \sin(k_y b), \quad (46)$$

$$A_A - A_B = -\lambda_{AM}(\mathbf{k}). \quad (47)$$

For $\mathbf{Q} = (h, k, l)$ in reciprocal-lattice units,

$$\lambda_{AM}(h, k, l) = 4S\delta J_7 \sin(2\pi h) \sin(2\pi k), \quad (48)$$

and

$$\lambda_{AM}(C_{4z}\mathbf{Q}) = -\lambda_{AM}(\mathbf{Q}). \quad (49)$$

5. Exchange BdG matrix from J_1 – J_7

All sublattice-even normal terms combine into

$$A_0(\mathbf{k}) = S \left\{ 2J_1(\cos z - 1) + 2J_3(\cos x + \cos y - 2) \right. \quad (50)$$

$$\left. + 4J_4[(\cos x + \cos y)\cos z - 2] + 2J_6(\cos 2z - 1) \right. \quad (51)$$

$$\left. + 4\bar{J}_7(\cos x \cos y - 1) + 8J_2 + 8J_5 \right\}. \quad (52)$$

The inter-sublattice pairing is

$$\Gamma(\mathbf{k}) = -8S \cos \frac{x}{2} \cos \frac{y}{2} \left(J_2 \cos \frac{z}{2} + J_5 \cos \frac{3z}{2} \right). \quad (53)$$

The exchange-only normal and pairing blocks are

$$A_{\text{ex}}(\mathbf{k}) = \begin{pmatrix} A_0 - \lambda_{\text{AM}}/2 & 0 \\ 0 & A_0 + \lambda_{\text{AM}}/2 \end{pmatrix}, \quad B_{\text{ex}}(\mathbf{k}) = \begin{pmatrix} 0 & \Gamma \\ \Gamma & 0 \end{pmatrix}. \quad (54)$$

In the Nambu basis

$$\Psi_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, a_{-\mathbf{k}}^\dagger, b_{-\mathbf{k}}^\dagger)^\top, \quad (55)$$

$$H_{\text{ex}}^{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} A_{\text{ex}}(\mathbf{k}) & B_{\text{ex}}(\mathbf{k}) \\ B_{\text{ex}}^\dagger(\mathbf{k}) & A_{\text{ex}}^*(\mathbf{k}) \end{pmatrix}. \quad (56)$$

With DDI, SIA, and field switched off, the common positive-frequency center is

$$\omega_0(\mathbf{k}) = \sqrt{A_0^2(\mathbf{k}) - \Gamma^2(\mathbf{k})}, \quad (57)$$

and the two positive frequencies can be written as

$$\omega_{\pm}(\mathbf{k}) = \omega_0(\mathbf{k}) \pm \frac{\lambda_{\text{AM}}(\mathbf{k})}{2}, \quad (58)$$

up to the overall branch-ordering convention. We take

$$\Lambda_{\mathbf{Q}} = -\lambda_{\text{AM}}(\mathbf{Q}). \quad (59)$$

6. Single-ion anisotropy in LSWT

6.1 Axial scalar SIA

For

$$\mathcal{H}_{\text{SIA}} = D_c \sum_i (S_i^z)^2, \quad (60)$$

we have locally

$$(\tilde{S}^z)^2 = (S - n)^2 = S^2 - 2Sn + O(S^0), \quad (61)$$

and therefore

$$A_{\text{SIA}}^{\text{axial}} = -2SD_c I_2, \quad B_{\text{SIA}}^{\text{axial}} = 0. \quad (62)$$

This corresponds to the code operation ‘A += -2*S*Dc’; the production model uses $D_c = 0$.

6.2 General quadratic SIA tensor

More generally,

$$\mathcal{H}_{\text{SIA}} = \sum_{i\mu} \mathbf{S}_{i\mu}^\top \mathbf{A}_\mu \mathbf{S}_{i\mu}, \quad \tilde{\mathbf{A}}_\mu = \mathbf{R}_\mu^\top \mathbf{A}_\mu \mathbf{R}_\mu. \quad (63)$$

At a stable ordered direction, the linear boson terms vanish. The quadratic contributions are

$$A_{\text{SIA},\mu} = S(\tilde{A}_{xx} + \tilde{A}_{yy} - 2\tilde{A}_{zz}), \quad (64)$$

$$B_{\text{SIA},\mu} = S \left[(\tilde{A}_{xx} - \tilde{A}_{yy}) + 2i\tilde{A}_{xy} \right], \quad (65)$$

The quadratic blocks are

$$A_{\text{SIA}} = \text{diag}(A_{\text{SIA},A}, A_{\text{SIA},B}), \quad B_{\text{SIA}} = \text{diag}(B_{\text{SIA},A}, B_{\text{SIA},B}). \quad (66)$$

For the microscopic SIA robustness check, the two ideal-*mmm* site tensors in a common crystallographic frame are approximately

$$\mathbf{A}_A = \text{diag}(-0.03, 0.53, -0.50) \mu\text{eV}, \quad (67)$$

$$\mathbf{A}_B = \text{diag}(0.53, -0.03, -0.50) \mu\text{eV}. \quad (68)$$

The two tensors are related by the x/y interchange between the C_{4z} -related sublattices and do not generate the same sublattice-odd diagonal detuning as δJ_7 . The numerical check gives a Γ gap of about 1.14345 meV for exchange+DDI and 1.16563 meV after adding this SIA; the main calculation uses the minimal exchange+DDI model.

7. Zeeman term and finite-field detuning

For $B \parallel c$,

$$\mathcal{H}_Z = -g\mu_B B \left(\sum_A S_A^z + \sum_B S_B^z \right). \quad (69)$$

Because $S_A^z = \tilde{S}_A^z$ and $S_B^z = -\tilde{S}_B^z$, the constants cancel and the quadratic term is

$$\mathcal{H}_Z^{(2)} = g\mu_B B \sum_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} - g\mu_B B \sum_{\mathbf{k}} b_{\mathbf{k}}^\dagger b_{\mathbf{k}}. \quad (70)$$

The normal block acquires

$$A_Z(B) = g\mu_B B \sigma_z, \quad (71)$$

and the signed energy detuning between the two diabatic modes changes by

$$h(B) = 2g\mu_B B. \quad (72)$$

8. Long-range DDI: from the real-space tensor to BdG

8.1 Momentum-space dipolar tensor

Define

$$\mathbb{T}(\mathbf{r}) = C_{\text{dip}} \left(\frac{I_3}{r^3} - 3 \frac{\mathbf{r}\mathbf{r}^T}{r^5} \right), \quad (73)$$

and

$$D_{\mu\nu}(\mathbf{Q}) = \sum_{\mathbf{R}}' \mathbb{T}(\mathbf{R} + \mathbf{r}_\nu - \mathbf{r}_\mu) e^{i\mathbf{Q} \cdot (\mathbf{R} + \mathbf{r}_\nu - \mathbf{r}_\mu)}. \quad (74)$$

The prime excludes the self pair $\mu = \nu$, $\mathbf{R} = 0$. Because the $1/r^3$ dipolar sum is conditionally convergent, the production calculation uses Ewald splitting.

8.2 Ewald decomposition

Let η be the Ewald parameter and $\mathbf{r} = \mathbf{R} + \mathbf{r}_\nu - \mathbf{r}_\mu$. The screened real-space tensor is

$$\mathbb{T}^{\text{real}}(\mathbf{r}) = C_{\text{dip}} \left[c_\delta(r) I_3 - c_{rr}(r) \mathbf{r}\mathbf{r}^T \right], \quad (75)$$

with

$$c_\delta(r) = \frac{\text{erfc}(\eta r)}{r^3} + \frac{2\eta}{\sqrt{\pi}} \frac{e^{-\eta^2 r^2}}{r^2}, \quad (76)$$

$$c_{rr}(r) = \frac{3 \text{erfc}(\eta r)}{r^5} + \frac{6\eta}{\sqrt{\pi}} \frac{e^{-\eta^2 r^2}}{r^4} + \frac{4\eta^3}{\sqrt{\pi}} \frac{e^{-\eta^2 r^2}}{r^2}. \quad (77)$$

The reciprocal-space part is

$$D_{\mu\nu}^{\text{recip}}(\mathbf{Q}) = C_{\text{dip}} \frac{4\pi}{V} \sum_{\mathbf{G}}' e^{-K^2/(4\eta^2)} \frac{\mathbf{K}\mathbf{K}^T}{K^2} e^{i\mathbf{G} \cdot (\mathbf{r}_\nu - \mathbf{r}_\mu)}, \quad \mathbf{K} = \mathbf{Q} + \mathbf{G}. \quad (78)$$

and the self term is

$$D_{\mu\nu}^{\text{self}} = -\delta_{\mu\nu} C_{\text{dip}} \frac{4\eta^3}{3\sqrt{\pi}} I_3. \quad (79)$$

The three contributions combine as

$$D_{\mu\nu}(\mathbf{Q}) = D_{\mu\nu}^{\text{real}}(\mathbf{Q}) + D_{\mu\nu}^{\text{recip}}(\mathbf{Q}) + D_{\mu\nu}^{\text{self}}. \quad (80)$$

The star excludes only terms with $\mathbf{K} = \mathbf{Q} + \mathbf{G} = 0$; no reciprocal term with $K \neq 0$ is discarded. The production parameters are $\eta = 0.15 \text{ \AA}^{-1}$, $R_{\text{real}} = 20 \text{ \AA}$, and $G_{\text{cut}} = 7 \text{ \AA}^{-1}$.

8.3 From the DDI tensor to A_{dip} and B_{dip}

Rotate each sublattice pair,

$$\tilde{D}_{\mu\nu}(\mathbf{Q}) = R_\mu^T D_{\mu\nu}(\mathbf{Q}) R_\nu. \quad (81)$$

The transverse normal and pairing coefficients are

$$t_{\mu\nu}^{\text{dip}}(\mathbf{Q}) = \frac{S}{2} \left[(\tilde{D}_{xx} + \tilde{D}_{yy}) + i(\tilde{D}_{yx} - \tilde{D}_{xy}) \right], \quad (82)$$

$$p_{\mu\nu}^{\text{dip}}(\mathbf{Q}) = \frac{S}{2} \left[(\tilde{D}_{xx} - \tilde{D}_{yy}) + i(\tilde{D}_{xy} + \tilde{D}_{yx}) \right]. \quad (83)$$

The corresponding BdG blocks are

$$[A_{\text{dip}}(\mathbf{Q})]_{\mu\nu} = t_{\mu\nu}^{\text{dip}}(\mathbf{Q}) - \delta_{\mu\nu} S \sum_{\rho} [\tilde{D}_{\mu\rho}(0)]_{zz}, \quad (84)$$

$$[B_{\text{dip}}(\mathbf{Q})]_{\mu\nu} = p_{\mu\nu}^{\text{dip}}(\mathbf{Q}). \quad (85)$$

The second term is the longitudinal restoring field about the static collinear state and is evaluated from $D(0)$. The complete DDI BdG contribution is

$$H_{\text{dip}}^{\text{BdG}}(\mathbf{Q}) = \begin{pmatrix} A_{\text{dip}}(\mathbf{Q}) & B_{\text{dip}}(\mathbf{Q}) \\ B_{\text{dip}}^{\dagger}(\mathbf{Q}) & A_{\text{dip}}^*(\mathbf{Q}) \end{pmatrix}. \quad (86)$$

All same-sublattice, inter-sublattice, normal, and anomalous blocks are retained; $D_{\mathbf{Q}}$ is not identified with a single AB tensor element.

9. Complete quadratic Hamiltonian and Colpa diagonalization

Combining all terms gives

$$A(\mathbf{Q}, B) = A_{\text{ex}} + A_{\text{SIA}} + A_Z + A_{\text{dip}}, \quad (87)$$

$$B(\mathbf{Q}) = B_{\text{ex}} + B_{\text{SIA}} + B_{\text{dip}}, \quad (88)$$

$$H^{\text{BdG}}(\mathbf{Q}, B) = \begin{pmatrix} A(\mathbf{Q}, B) & B(\mathbf{Q}) \\ B^{\dagger}(\mathbf{Q}) & A^*(\mathbf{Q}, B) \end{pmatrix}. \quad (89)$$

The quadratic bosonic Hamiltonian is

$$\mathcal{H}^{(2)} = E_{\text{cl}} + \frac{1}{2} \sum_{\mathbf{Q}} \Psi_{\mathbf{Q}}^{\dagger} H^{\text{BdG}}(\mathbf{Q}) \Psi_{\mathbf{Q}} + \text{const.} \quad (90)$$

The bosonic eigenproblem has the generalized form

$$H^{\text{BdG}} u_n = \omega_n \eta u_n, \quad \eta = \text{diag}(1, 1, -1, -1), \quad (91)$$

or equivalently one diagonalizes ηH^{BdG} . The production calculation uses a Colpa/Cholesky procedure and enforces the paraunitary condition

$$T^{\dagger} \eta T = \eta. \quad (92)$$

The paraunitary error is monitored by

$$\epsilon_{\text{para}} = \max |T^{\dagger} \eta T - \eta|. \quad (93)$$

10. Effective two-mode Hamiltonian: projection from 4×4 BdG to 2×2

Near the target momentum, the full 4×4 bosonic BdG problem contains an isolated two-dimensional positive-frequency subspace. Projecting the full Hamiltonian onto this subspace gives the general 2×2 Hermitian effective Hamiltonian

$$H_{\text{eff}} = \bar{\omega} I + \frac{1}{2} (\Lambda + h) \sigma_z + d_x \sigma_x + d_y \sigma_y, \quad (94)$$

whose eigenvalue splitting satisfies

$$\Omega^2 = (\Lambda + h)^2 + 4(d_x^2 + d_y^2). \quad (95)$$

The exchange imbalance selects the diabatic basis and fixes the physical meaning of the Pauli components; the exchange, Zeeman, and DDI projections follow.

10.1 Equal- J_7 exchange-only reference doublet

Set

$$J_{7a} = J_{7b} = \bar{J}_7, \quad (96)$$

with DDI, SIA, and field switched off, so that $\lambda_{\text{AM}} = 0$. The exchange BdG matrix is then

$$H_0(\mathbf{k}) = \begin{pmatrix} A_0 & 0 & 0 & \Gamma \\ 0 & A_0 & \Gamma & 0 \\ 0 & \Gamma & A_0 & 0 \\ \Gamma & 0 & 0 & A_0 \end{pmatrix}, \quad \eta = \text{diag}(1, 1, -1, -1). \quad (97)$$

The generalized eigenproblem

$$H_0 u = \omega \eta u \quad (98)$$

decomposes into two identical 2×2 bosonic blocks, $(a_{\mathbf{k}}, b_{-\mathbf{k}}^{\dagger})$ and $(b_{\mathbf{k}}, a_{-\mathbf{k}}^{\dagger})$. The two positive-frequency branches are therefore exactly degenerate,

$$\omega_0(\mathbf{k}) = \sqrt{A_0^2(\mathbf{k}) - \Gamma^2(\mathbf{k})}. \quad (99)$$

A positive-norm basis may be chosen as

$$u_1 = \begin{pmatrix} u \\ 0 \\ 0 \\ v \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 \\ u \\ v \\ 0 \end{pmatrix}, \quad (100)$$

with

$$u^2 - v^2 = 1, \quad v = -\frac{\Gamma}{A_0 + \omega_0} u. \quad (101)$$

Equivalently,

$$u = \sqrt{\frac{1}{2} \left(\frac{A_0}{\omega_0} + 1 \right)}, \quad |v| = \sqrt{\frac{1}{2} \left(\frac{A_0}{\omega_0} - 1 \right)}, \quad (102)$$

with the sign of v fixed by Γ . Defining

$$U_0 = (u_1, u_2), \quad U_0^\dagger \eta U_0 = I_2, \quad (103)$$

makes the target positive-frequency subspace two-dimensional, and the two-mode effective model is the projection of the full BdG problem onto this subspace.

10.2 Projection in the bosonic BdG subspace

Let

$$H = H_0 + \delta H, \quad \omega = \omega_0 + \delta \omega, \quad u = U_0 c + \dots \quad (104)$$

Substitution into

$$Hu = \omega \eta u \quad (105)$$

and left multiplication by $U_0^\dagger$, using

$$H_0 U_0 = \omega_0 \eta U_0, \quad U_0^\dagger \eta U_0 = I_2, \quad (106)$$

gives, to first order,

$$U_0^\dagger \delta H U_0 c = \delta \omega c. \quad (107)$$

The first-order effective operator in the degenerate positive-frequency doublet is

$$M[\delta H] = U_0^\dagger \delta H U_0. \quad (108)$$

The positive-norm normalization already includes the bosonic metric η , so no additional η enters the projected matrix.

10.3 Exchange-imbalance contribution

Restoring the exchange imbalance introduces the normal-block term

$$\delta A_{\text{AM}} = \begin{pmatrix} -\lambda/2 & 0 \\ 0 & +\lambda/2 \end{pmatrix}. \quad (109)$$

In the full Nambu space,

$$\delta H_{\text{AM}} = \text{diag}(-\lambda/2, +\lambda/2, -\lambda/2, +\lambda/2). \quad (110)$$

Projection onto u_1, u_2 gives

$$\langle u_1 | \delta H_{\text{AM}} | u_1 \rangle = -\frac{\lambda}{2} u^2 + \frac{\lambda}{2} v^2 = -\frac{\lambda}{2} (u^2 - v^2) = -\frac{\lambda}{2}, \quad (111)$$

$$\langle u_2 | \delta H_{\text{AM}} | u_2 \rangle = +\frac{\lambda}{2} u^2 - \frac{\lambda}{2} v^2 = +\frac{\lambda}{2}, \quad (112)$$

$$\langle u_1 | \delta H_{\text{AM}} | u_2 \rangle = 0. \quad (113)$$

The projection gives

$$M_{\text{AM}} = -\frac{\lambda}{2} \sigma_z = \frac{\Lambda}{2} \sigma_z, \quad \Lambda \equiv -\lambda. \quad (114)$$

In the chosen branch ordering, $\Lambda = -\lambda_{\text{AM}}$ is the signed diagonal detuning generated by the exchange imbalance in the positive-frequency doublet.

10.4 Zeeman contribution

The quadratic Zeeman matrix is

$$\delta H_Z = g\mu_B B \text{ diag}(+1, -1, +1, -1). \quad (115)$$

Projection gives

$$\langle u_1 | \delta H_Z | u_1 \rangle = +g\mu_B B(u^2 - v^2) = +g\mu_B B, \quad (116)$$

$$\langle u_2 | \delta H_Z | u_2 \rangle = -g\mu_B B(u^2 - v^2) = -g\mu_B B, \quad (117)$$

$$\langle u_1 | \delta H_Z | u_2 \rangle = 0. \quad (118)$$

The projection gives

$$M_Z = g\mu_B B \sigma_z. \quad (119)$$

The relative diabatic detuning is

$$\Lambda + h(B) = \Lambda + 2g\mu_B B. \quad (120)$$

The Zeeman and exchange detunings act along the same Pauli direction in the same two-dimensional Hilbert space; compensation occurs when their diagonal contributions cancel.

10.5 Projected dipolar mixing

Projecting the complete Ewald DDI BdG operator into the same diabatic basis gives

$$M_{\text{DDI}} = U^\dagger H_{\text{dip}}^{\text{BdG}} U. \quad (121)$$

As a 2×2 Hermitian matrix, M_{DDI} has the unique decomposition

$$M_{\text{DDI}} = d_0 I + d_x \sigma_x + d_y \sigma_y + d_z \sigma_z. \quad (122)$$

Explicitly, if

$$M_{\text{DDI}} = \begin{pmatrix} m_{11} & V \\ V^* & m_{22} \end{pmatrix}, \quad (123)$$

then

$$d_0 = \frac{m_{11} + m_{22}}{2}, \quad (124)$$

$$d_z = \frac{m_{11} - m_{22}}{2}, \quad (125)$$

$$d_x = \Re V, \quad (126)$$

$$d_y = -\Im V. \quad (127)$$

The off-diagonal part defines the transverse mixing,

$$D_Q = 2|V_Q| = 2\sqrt{d_x^2 + d_y^2}. \quad (128)$$

Explicit projection gives $2d_z \simeq 0$ to machine precision, so the DDI contributes only an identity shift and transverse mixing in this basis. A finite d_z can be absorbed into the total diagonal detuning.

10.6 Two-mode eigenvalue relation

Combining the projected terms and absorbing common energy shifts into $\bar{\omega}$ gives

$$H_{\text{eff}} = \bar{\omega} I + \begin{pmatrix} (\Lambda + h + 2d_z)/2 & V \\ V^* & -(\Lambda + h + 2d_z)/2 \end{pmatrix}. \quad (129)$$

For a matrix

$$\begin{pmatrix} z & V \\ V^* & -z \end{pmatrix}, \quad (130)$$

the eigenvalues are

$$\pm \sqrt{z^2 + |V|^2}. \quad (131)$$

The two-level splitting is

$$\Omega = 2\sqrt{\left(\frac{\Lambda + h + 2d_z}{2}\right)^2 + |V|^2}, \quad (132)$$

or

$$\Omega^2 = (\Lambda + h + 2d_z)^2 + D_Q^2. \quad (133)$$

Using the result $2d_z \simeq 0$ above and $h = 2g_{\text{eff}}\mu_B B$ gives

$$\Omega_Q^2(B) = [\Lambda_Q + 2g_{\text{eff}}\mu_B B]^2 + D_Q^2. \quad (134)$$

The sum-of-squares relation follows directly from the eigenvalues of the 2×2 Hermitian effective Hamiltonian. It applies when the target doublet is separated from the remaining positive- and negative-frequency sectors and the Pauli components are identified in the exchange-defined diabatic basis.

11. From the full BdG problem to the target doublet: $\Lambda_{\mathbf{Q}}$ and $D_{\mathbf{Q}}$

11.1 Exchange-defined reference doublet

To distinguish the full-BdG gap from the projected DDI mixing parameter, define the symmetric exchange-only reference

$$J_{7a} = J_{7b} = \bar{J}_7, \quad (135)$$

with DDI and field switched off while all other exchange couplings are kept unchanged:

$$H_0(\mathbf{Q}) = H_{\text{ex}}^{\text{BdG}}(\mathbf{Q}; J_{7a} = J_{7b} = \bar{J}_7). \quad (136)$$

At the target momentum, the two positive-frequency reference modes are numerically degenerate. Let

$$U_0 = (u_1, u_2), \quad U_0^\dagger \eta U_0 = I_2. \quad (137)$$

For a small BdG perturbation δH , the projected matrix in the degenerate positive-frequency subspace is

$$M[\delta H] = U_0^\dagger \delta H U_0. \quad (138)$$

This uses the positive-norm projection rule established above.

11.2 Fixing the diabatic basis with the exchange imbalance

Define

$$\delta H_{\text{AM}}(\mathbf{Q}) = H_{\text{ex}}^{\text{split}}(\mathbf{Q}) - H_0(\mathbf{Q}). \quad (139)$$

The exchange-imbalance projection is

$$M_{\text{AM}} = U_0^\dagger \delta H_{\text{AM}} U_0. \quad (140)$$

Because H_0 is degenerate, arbitrary linear combinations of u_1, u_2 are allowed. Diagonalizing the 2×2 Hermitian matrix M_{AM} fixes the diabatic basis,

$$U = U_0 W, \quad (141)$$

with the two diabatic branches ordered according to the convention

$$\Lambda_{\mathbf{Q}} = -\lambda_{\text{AM}}(\mathbf{Q}). \quad (142)$$

Then

$$U^\dagger \delta H_{\text{AM}} U = \begin{pmatrix} \Lambda_{\mathbf{Q}}/2 & 0 \\ 0 & -\Lambda_{\mathbf{Q}}/2 \end{pmatrix} \quad (143)$$

up to an identity shift that does not affect the splitting.

11.3 Explicit projection of the complete DDI

In the same exchange-defined basis, the Ewald DDI operator projects to

$$M_{\text{DDI}}(\mathbf{Q}) = U^\dagger H_{\text{dip}}^{\text{BdG}}(\mathbf{Q}) U. \quad (144)$$

Using the Pauli decomposition above,

$$M_{\text{DDI}} = d_0 I + d_x \sigma_x + d_y \sigma_y + d_z \sigma_z. \quad (145)$$

If

$$V_{\mathbf{Q}} = [M_{\text{DDI}}]_{12} = d_x - i d_y, \quad (146)$$

then

$$D_{\mathbf{Q}} = 2|V_{\mathbf{Q}}| = 2\sqrt{d_x^2 + d_y^2}. \quad (147)$$

The projected diagonal DDI detuning is

$$\Lambda_{\text{DDI}}^{\text{diag}} = 2d_z \quad (148)$$

and is evaluated explicitly at the momentum of interest.

The individual d_x, d_y depend on the phases chosen for the two diabatic basis vectors. Under

$$u_1 \rightarrow e^{i\phi_1} u_1, \quad u_2 \rightarrow e^{i\phi_2} u_2, \quad (149)$$

$V_{\mathbf{Q}}$ acquires a phase, whereas

$$D_{\mathbf{Q}} = 2|V_{\mathbf{Q}}| \quad (150)$$

remains invariant. Hence $D_{\mathbf{Q}}$ is the phase-gauge-invariant mixing scale, whereas d_x and d_y separately have no independent physical meaning.

11.4 Numerical validation of the projected parameter

At the working momentum $\mathbf{Q}_A = (0.25, 0.75, 0)$, explicit projection gives

$$D_A^{\text{proj}} = 121.809730 \mu\text{eV}, \quad (151)$$

whereas the equal- J_7 full-BdG avoided-crossing floor including the complete DDI is

$$\Omega_{\text{eq}}^{\text{full}}(0) = 121.592033 \mu\text{eV}. \quad (152)$$

The difference is $0.217697 \mu\text{eV}$, or 0.179% , which measures higher-order corrections from Nambu sectors outside the target doublet; the projected parameter and the complete full-BdG floor differ at the sub-percent level.

12. Two-mode formula and compensation field

Neglecting the projected diagonal DDI correction, the target doublet is described by

$$H_{\text{eff}}(\mathbf{Q}, B) = \bar{\omega}_{\mathbf{Q}} I + \frac{1}{2} [\Lambda_{\mathbf{Q}} + 2g_{\text{eff}}\mu_B B] \sigma_z + d_x \sigma_x + d_y \sigma_y. \quad (153)$$

The eigenvalue splitting satisfies

$$\Omega_{\mathbf{Q}}^2(B) = [\Lambda_{\mathbf{Q}} + 2g_{\text{eff}}\mu_B B]^2 + D_{\mathbf{Q}}^2. \quad (154)$$

If a diagonal DDI correction is retained, replace $\Lambda_{\mathbf{Q}}$ by

$$\Lambda_{\mathbf{Q}}^{\text{tot}} = \Lambda_{\mathbf{Q}} + 2d_z. \quad (155)$$

The minimum splitting occurs at

$$B^*(\mathbf{Q}) = -\frac{\Lambda_{\mathbf{Q}}}{2g_{\text{eff}}\mu_B}. \quad (156)$$

Here $D_{\mathbf{Q}}$ fixes the minimum avoided-crossing gap, while the symmetry-odd exchange detuning fixes its position along the field axis.

At zero field,

$$\Omega_{\mathbf{Q}}(0) = \sqrt{\Lambda_{\mathbf{Q}}^2 + D_{\mathbf{Q}}^2}. \quad (157)$$

For $|\Lambda| \ll D$,

$$\Omega(0) - D \simeq \frac{\Lambda^2}{2D}, \quad (158)$$

The weak exchange enters the zero-field splitting in an even, quadratic form, whereas B^* is linear in Λ and signed.

13. Symmetry-related momentum partners and fixed-field contrast

Take

$$\mathbf{Q}_A = (0.25, 0.75, 0), \quad \mathbf{Q}_{A'} = (-0.75, 0.25, 0) = C_{4z} \mathbf{Q}_A. \quad (159)$$

From

$$\lambda_{\text{AM}}(\mathbf{Q}_A) = -4S\delta J_7, \quad \lambda_{\text{AM}}(\mathbf{Q}_{A'}) = +4S\delta J_7, \quad (160)$$

we obtain

$$\Lambda_A = +4S\delta J_7, \quad \Lambda_{A'} = -4S\delta J_7. \quad (161)$$

The full-BdG symmetry gives

$$\Omega_{A'}(B) = \Omega_A(-B), \quad (162)$$

and hence

$$B_A^* = -B_{A'}^* = -\frac{2S\delta J_7}{g_{\text{eff}}\mu_B}. \quad (163)$$

Define

$$C(B) = \Omega_A^2(B) - \Omega_{A'}^2(B). \quad (164)$$

If the projected partner mixing satisfies $D_A^2 = D_{A'}^2$, the common mixing term cancels, giving

$$C(B) = 8g_{\text{eff}}\mu_B \Lambda_A B, \quad (165)$$

with slope

$$\frac{dC}{dB} = 8g_{\text{eff}}\mu_B \Lambda_A = 32Sg_{\text{eff}}\mu_B \delta J_7. \quad (166)$$

14. Conversion of D_c in the FeF_2 transfer calculation

The FeF_2 literature model uses the axial single-ion term

$$\mathcal{H}_{\text{SIA}}^{\text{lit}} = -D \sum_i (S_i^z)^2, \quad D = 0.82 \text{ meV}, \quad S = 2. \quad (167)$$

The code uses the convention

$$\mathcal{H}_{\text{SIA}}^{\text{code}} = D_c \sum_i (S_i^z)^2, \quad (168)$$

with harmonic HP contribution

$$A_{\text{SIA}}^{\text{axial}} = -2SD_c. \quad (169)$$

The Hamiltonian sign convention gives $D_c = -D$. To reproduce the spin-wave energy scale associated with the published FeF_2 value $D = 0.82 \text{ meV}$ in the LSWT/BdG convention, we use the finite- S conversion

$$D_c^{\text{core}} = -D \left(1 - \frac{1}{2S}\right). \quad (170)$$

For $S = 2$,

$$D_c^{\text{core}} = -0.82 \left(1 - \frac{1}{4}\right) = -0.615 \text{ meV}. \quad (171)$$

The factor follows from the finite- S level spacing of a single spin. For $-D(S^z)^2$, the exact excitation energy from $m = S$ to $m = S - 1$ is

$$\Delta E_{S \rightarrow S-1} = D [S^2 - (S-1)^2] = D(2S-1). \quad (172)$$

The harmonic HP coefficient has scale $2S|D_c^{\text{core}}|$. Equating the two gives

$$2S|D_c^{\text{core}}| = D(2S-1), \quad (173)$$

and hence

$$|D_c^{\text{core}}| = D \left(1 - \frac{1}{2S}\right). \quad (174)$$

The factor $1 - 1/(2S)$ maps the finite- S single-ion level spacing onto the harmonic bosonic coefficient and is used only as a convention conversion, without introducing an additional interaction or fitted parameter.